

Brain Tumor Segmentation with Swin Transformers

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Code: <https://github.com/ducdonghiem/swin-brats-domain-shift>

Abstract—We investigate the adaptation of the Swin Transformer, a hierarchical vision Transformer originally designed for natural image understanding, to the task of multi-modal brain tumor segmentation on the BraTS 2021 dataset. The domain shift from three-channel RGB natural images to four-channel multi-parametric MRI data introduces significant architectural and data-level challenges, including a change in input dimensionality, severe class imbalance, and a fundamentally different visual distribution. We propose SwinBraTS, a U-Net-style encoder-decoder built on a scaled-down Swin-T backbone, augmented with a modality projection and reconstruction block, and skip connections to preserve spatial context. We evaluate performance using the Dice Similarity Coefficient (DSC) and 95th percentile Hausdorff Distance (HD95). We further conduct an ablation study by varying the base channel width parameter C to characterize model sensitivity under limited training data. Our results demonstrate that SwinBraTS achieves a mean DSC of 0.6645 on the test set, with per-region scores of 0.7761 on the Whole

Tumor (WT), 0.6788 on the Tumor Core (TC), and 0.5384 on the Enhancing Tumor (ET), demonstrating that the hierarchical shifted-window attention mechanism transfers meaningfully to multi-modal medical imaging despite the substantial domain gap from its natural image origins.

Index Terms—Swin Transformer, brain tumor segmentation, BraTS 2021, domain shift, medical image segmentation, multi-modal MRI, U-Net, ablation study

I. INTRODUCTION

Accurate segmentation of brain tumors from magnetic resonance imaging (MRI) is a critical step in diagnosis, surgical planning, and treatment monitoring. Gliomas, among the most aggressive primary brain tumors, present subregions that vary substantially in size, shape, and imaging contrast, requiring clinicians to examine multiple MRI modalities simultaneously [1–4]. Manual analysis is time-consuming and subject to inter-rater variability, mo-

tivating the development of automated segmentation pipelines.

Convolutional neural networks (CNNs), and in particular U-Net [5] and its variants, have dominated medical image segmentation for several years. More recently, the Vision Transformer (ViT) [6] and its successors have demonstrated competitive performance on natural image benchmarks. The Swin Transformer [7], published at ICCV 2021, addresses key limitations of ViT by introducing a hierarchical architecture with shifted window self-attention, enabling linear computational complexity and multi-scale feature extraction. Swin achieved state-of-the-art results on ImageNet1K [8] classification, COCO [9] object detection, and ADE20K [10] semantic segmentation, all natural image tasks.

The question we investigate is: *how well does the Swin Transformer’s hierarchical windowed attention transfer to the domain of multi-modal medical imaging?* This constitutes a challenging domain shift. Unlike the three-channel RGB inputs Swin was designed for, BraTS 2021 [1] provides four MRI modalities (T1, T1Gd, T2, T2-FLAIR), each encoding distinct physiological information. The visual statistics, spatial structure, and class distribution differ fundamentally from natural images.

Our specific contribution is the design and evaluation of **SwinBraTS**, an adaptation of the Swin-T backbone to 3D multi-modal brain tumor segmenta-

tion, incorporating: (i) a modality projection block to bridge the 4-to-3 channel mismatch, (ii) U-Net-style skip connections for spatial context recovery, (iii) a reconstruction block with transposed convolutions, and (iv) a scaled-down model capacity suited to the BraTS dataset size. We additionally conduct an ablation study on the base embedding dimension C to understand how model capacity interacts with limited medical training data.

II. METHODOLOGY AND ADAPTATION

We propose **SwinBraTS**, a 3D segmentation model built by adapting the Swin-T backbone [7] to multi-modal MRI. The adaptation required four principal modifications to the original design: (i) a modality projection block to handle four input channels, (ii) a U-Net-style decoder with skip connections, (iii) a reconstruction block for dense prediction, and (iv) a scaled-down model capacity. We describe the original architecture first, then each modification in turn.

A. Original Swin Transformer Architecture

The Swin Transformer [7] is a hierarchical vision Transformer designed to serve as a general-purpose backbone for computer vision tasks. Given an input RGB image of size $H \times W \times 3$, the architecture begins with a patch partition module that splits the image into non-overlapping 4×4 patches, producing $\frac{H}{4} \times \frac{W}{4}$ tokens each with a raw feature dimension

of 48. A linear embedding layer then projects these features to an arbitrary hidden dimension C .

The architecture is organized into four successive stages. Stage 1 applies Swin Transformer blocks at $\frac{H}{4} \times \frac{W}{4}$ resolution. Stages 2, 3, and 4 each begin with a patch merging layer that concatenates the features of 2×2 neighboring patches and applies a linear projection, halving the spatial resolution and doubling the channel dimension. This produces a feature hierarchy at $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, and $\frac{1}{32}$ of the input resolution, analogous to the multi-scale feature maps of convolutional backbones such as ResNet [11].

Each Swin Transformer block replaces standard global multi-head self-attention (MSA) with a shifted window-based variant. In alternating blocks, self-attention is first computed within fixed non-overlapping local windows of size $M \times M$ patches (W-MSA), then within windows shifted by $(\lfloor \frac{M}{2} \rfloor, \lfloor \frac{M}{2} \rfloor)$ pixels (SW-MSA). This shift introduces cross-window connections without sacrificing computational efficiency, reducing the complexity of global self-attention from $\mathcal{O}((HW)^2)$ to $\mathcal{O}(HW)$ for a fixed window size. Relative position biases are added to the attention computation to encode spatial relationships between patches within each window.

The Swin-T variant (Swin-Tiny), uses $C = 96$ with layer counts $\{2, 2, 6, 2\}$ across the four stages, yielding approximately 28M parameters. It was

originally trained and evaluated on natural image datasets including ImageNet-1K [8] for classification, COCO [9] for object detection, and ADE20K [10] for semantic segmentation, all involving three-channel RGB inputs, which motivates the domain adaptation described in the following subsections. The full architecture is illustrated in Fig. 1.

B. SwinBraTS Overview

SwinBraTS is a U-Net-style encoder-decoder network built on a Swin Transformer backbone, designed to bridge the domain gap between natural image and multi-modal volumetric MRI segmentation. Fig. 2 illustrates an overview of the complete pipeline.

The architecture comprises four sequential stages. First, the **Projection Block** (Section II-D) accepts the four MRI modalities—each of spatial resolution 240×240 with depth $D = 155$ —and produces a single three-channel 224×224 representation compatible with the Swin backbone. This stage is entirely convolutional.

Second, the **Swin Encoder** (Section II-E) processes this representation through three hierarchical stages of Swin Transformer Blocks (Section II-C), progressively halving the spatial resolution and doubling the channel width, producing a multi-scale feature hierarchy at $\frac{1}{4}$, $\frac{1}{8}$, and $\frac{1}{16}$ of the input resolution. Skip tensors are retained at each scale for later use in the decoder.

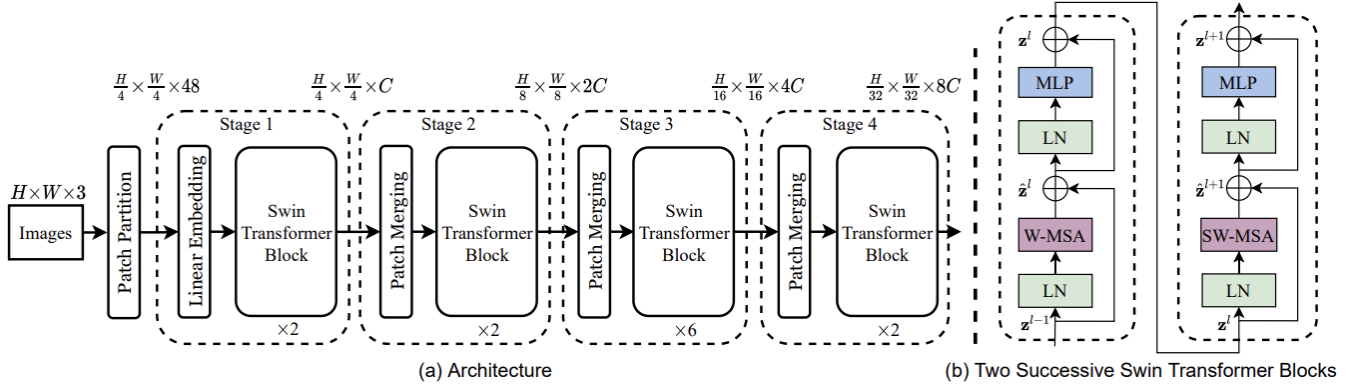


Fig. 1. Architecture of the Swin-T backbone [7]. (a) The four-stage hierarchical encoder with patch partition, linear embedding, and patch merging layers producing feature maps at $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$, and $\frac{1}{32}$ of input resolution. (b) Two successive Swin Transformer blocks alternating between window-based self-attention (W-MSA) and shifted window self-attention (SW-MSA).

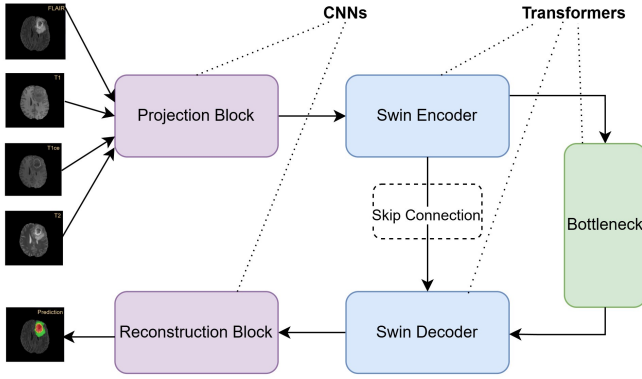


Fig. 2. Overall architecture of SwinBraTS. Four MRI modalities (T2-FLAIR, T1, T1Gd, T2) are fused by the CNN-based Projection Block into a three-channel representation, which is processed by the Swin Encoder-Bottleneck-Decoder backbone. Skip connections transfer spatial detail from encoder to decoder at each scale. The CNN-based Reconstruction Block maps the decoder output back to a full volumetric segmentation.

Third, the **Bottleneck** (Section II-F) operates at the coarsest resolution of 7×7 , where windowed attention degenerates to global self-attention over all 49 tokens, enabling the model to capture long-range spatial dependencies before decoding begins.

Fourth, the **Swin Decoder** (Section II-G) mirrors the encoder with three upsampling stages. At each scale, encoder skip features are fused with the

upsampled decoder features via a dedicated Skip Connection module before the Swin Transformer Blocks refine the combined representation. A Final Patch Expanding layer restores the spatial resolution to 224×224 .

Finally, the **Reconstruction Block** (Section II-H) maps the two-dimensional decoder output back to a full three-dimensional segmentation logit tensor of shape (B, K, D, H, W) , where $K = 4$ denotes the number of tumor subregion classes.

The overall design reflects two deliberate adaptations to the medical imaging domain. The convolutional projection and reconstruction stages handle the 4-to-3 channel mismatch and the 2D-to-3D volume reconstruction respectively, while the transformer backbone operates entirely in the standardised 224×224 token space for which the Swin architecture was designed.

C. Swin Transformer Block

The Swin Transformer Block (Fig. 3) is the fundamental computational unit shared across the encoder, bottleneck, and decoder of SwinBraTS. Each block takes a sequence of patch tokens $\mathbf{x} \in \mathbb{R}^{B \times HW \times C}$ and applies layer normalisation, window-based multi-head self-attention (W-MSA or SW-MSA), a residual connection, and a feed-forward network (FFN):

$$\hat{\mathbf{x}}^l = \text{W-MSA}(\text{LN}(\mathbf{x}^{l-1})) + \mathbf{x}^{l-1}, \quad (1)$$

$$\mathbf{x}^l = \text{FFN}(\text{LN}(\hat{\mathbf{x}}^l)) + \hat{\mathbf{x}}^l. \quad (2)$$

In alternating blocks the attention variant switches to SW-MSA, which shifts the partition grid by $(\lfloor M/2 \rfloor, \lfloor M/2 \rfloor)$ tokens to introduce cross-window interactions without increasing complexity.

Window Attention: Given a feature map of spatial resolution $H \times W$, tokens are partitioned into non-overlapping windows of size $M \times M$ (here $M = 7$), yielding $\lceil H/M \rceil \times \lceil W/M \rceil$ windows each containing $M^2 = 49$ tokens. Self-attention is computed independently within each window. For a window token sequence $\mathbf{z} \in \mathbb{R}^{M^2 \times C}$, the attention output is

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}} + \mathbf{B}\right) \mathbf{V}, \quad (3)$$

where $d_k = C/n_h$ is the per-head dimension, n_h is the number of attention heads, and $\mathbf{B} \in \mathbb{R}^{M^2 \times M^2}$ is a learnable relative position bias indexed by a pre-

computed position table of size $(2M - 1)^2 \times n_h$.

Cyclic Shift and Attention Mask: For SW-MSA, the feature map is cyclically shifted by $(-\lfloor M/2 \rfloor, -\lfloor M/2 \rfloor)$ before windowing. An additive attention mask of value -100 is applied to token pairs that are not spatially adjacent in the original (unshifted) map, suppressing cross-region attention after the softmax. The shift is reversed after attention. This mechanism introduces cross-window connectivity at zero additional computational cost.

Feed-Forward Network: The FFN is a two-layer MLP with GELU activation and an expansion ratio of $r = 4$: $\text{FFN}(\mathbf{x}) = \mathbf{W}_2 \text{GELU}(\mathbf{W}_1 \mathbf{x})$, where $\mathbf{W}_1 \in \mathbb{R}^{C \times 4C}$ and $\mathbf{W}_2 \in \mathbb{R}^{4C \times C}$. Stochastic depth (DropPath) [12] with linearly increasing rate is applied to the residual branches during training to regularise the deeper stages.

Each encoder and decoder stage consists of two consecutive Swin Transformer Blocks—one W-MSA and one SW-MSA—following the original Swin-T design. The bottleneck uses the same pair at the lowest spatial resolution of 7×7 .

D. Modality Projection Block

BraTS 2021 provides four MRI modalities (T1, T1Gd, T2, and T2-FLAIR), each a volumetric scan of spatial size $240 \times 240 \times 155$. The Swin backbone expects a three-channel 2D image of size 224×224 . The Projection Block (Fig. 4) bridges this gap in two steps.

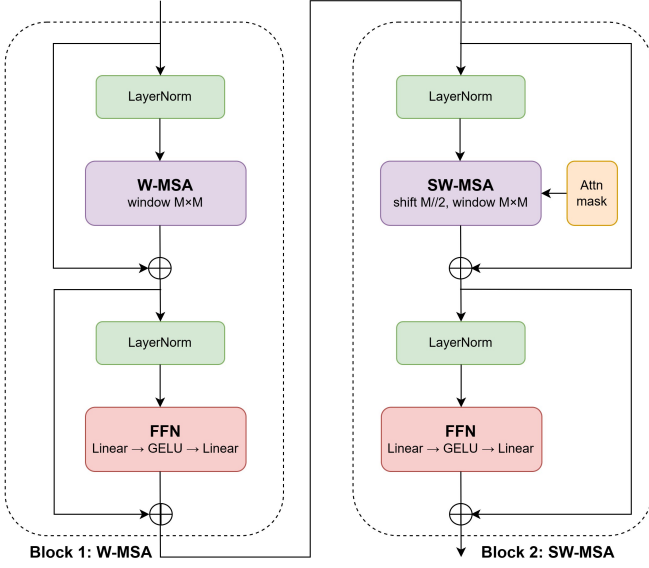


Fig. 3. A pair of consecutive Swin Transformer Blocks. The first block uses standard window attention (W-MSA); the second uses shifted window attention (SW-MSA). FFN denotes the two-layer feed-forward network.

Per-Modality CNN: Each modality is first treated as a 2D tensor by interpreting the depth dimension $D = 155$ as input channels, giving a representation $\mathbf{m}_i \in \mathbb{R}^{B \times 155 \times 240 \times 240}$. A dedicated convolutional branch f_i , which does not share weights across modalities, extracts spatial features while simultaneously compressing depth:

$$\begin{aligned} \mathbf{h}_i &= f_i(\mathbf{m}_i), \\ f_i &: \mathbb{R}^{B \times 155 \times 240 \times 240} \rightarrow \mathbb{R}^{B \times H_p \times 224 \times 224}, \end{aligned} \quad (4)$$

where H_p is the projection hidden dimension (16 in our experiments). Each branch f_i consists of three consecutive convolutions with kernel sizes 11, 5, and 3, reducing the spatial resolution from 240×240 to 224×224 while building a multi-scale feature representation. Specifically, the spatial

progression is $240 \xrightarrow{11 \times 11} 230 \xrightarrow{5 \times 5} 226 \xrightarrow{3 \times 3} 224$.

GroupNorm (with $G = 1$, equivalent to instance normalisation) and GELU activation follow each convolution. BatchNorm is deliberately avoided: with the small batch sizes required by 3D medical volumes (typically $B = 4$), batch statistics are insufficiently stable.

Cross-Modality Fusion: The four branch outputs are concatenated along the channel axis, yielding $\mathbf{H} \in \mathbb{R}^{B \times 4H_p \times 224 \times 224}$, and fused by a 1×1 convolution followed by GroupNorm:

$$\mathbf{F} = \text{GN}(\mathbf{W}_f * [\mathbf{h}_1 \parallel \mathbf{h}_2 \parallel \mathbf{h}_3 \parallel \mathbf{h}_4]) \in \mathbb{R}^{B \times 3 \times 224 \times 224}. \quad (5)$$

No activation is applied after fusion: the Swin Patch Partition normalises its inputs internally, and a pre-fusion ReLU would eliminate all negative activations, discarding roughly half the signal from zero-centred normalised MRI intensities.

E. Swin Encoder

The encoder (Fig. 5) follows the Swin-T hierarchy. The projected three-channel input $\mathbf{F} \in \mathbb{R}^{B \times 3 \times 224 \times 224}$ is first processed by a Patch Partition module: a 4×4 strided convolution embeds non-overlapping patches into tokens of dimension 48, producing a sequence of 56×56 tokens. A linear layer projects these to the base channel width C :

$$\mathbf{X}_0 \in \mathbb{R}^{B \times 3136 \times C}. \quad (6)$$

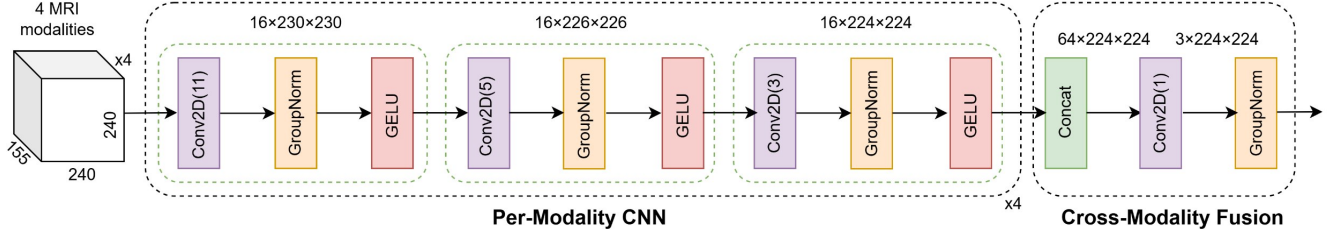


Fig. 4. The Modality Projection Block. Four independent CNN branches extract per-modality spatial features from depth-as-channel representations; their outputs are concatenated and fused to a three-channel 224×224 tensor compatible with the Swin encoder.

Three encoder stages then process the token sequence at progressively lower resolutions. Between stages, Patch Merging halves the spatial dimensions by concatenating 2×2 neighbouring tokens and applying a normalised linear projection:

$$\text{PatchMerge} : \mathbb{R}^{B \times HW \times C} \rightarrow \mathbb{R}^{B \times \frac{HW}{4} \times 2C}. \quad (7)$$

The resulting feature hierarchy, with dimensions and attention heads summarised in Table I, produces three skip tensors $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$ and a bottleneck input $\mathbf{E}_b$ at 7×7 resolution.

TABLE I
ENCODER STAGE CONFIGURATION ($C = 96$).

Stage	Res.	Ch.	Heads
Patch Partition + Proj.	56^2	C	–
Encoder Stage 1	56^2	C	3
Encoder Stage 2	28^2	$2C$	6
Encoder Stage 3	14^2	$4C$	12
Patch Merging	7^2	$8C$	–

F. Bottleneck

The bottleneck (Fig. 6) processes the coarsest representation $\mathbf{E}_b \in \mathbb{R}^{B \times 49 \times 8C}$ at spatial resolution 7×7 . It consists of two Swin Transformer Blocks

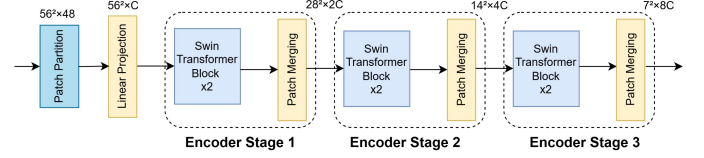


Fig. 5. The Swin encoder. Patch Partition and a linear projection initialise the token sequence. Three stages of Swin Transformer Blocks interleaved with Patch Merging layers build a four-scale feature hierarchy. Skip tensors $\mathbf{E}_1$, $\mathbf{E}_2$, $\mathbf{E}_3$ are forwarded to the corresponding decoder stages.

(one W-MSA, one SW-MSA) with $n_h = 24$ attention heads. At this resolution the 7×7 feature map fits within a single window, so windowed attention degenerates to global attention over all 49 tokens. This gives the model unrestricted global receptive field at the lowest scale, analogous to the bridge in a convolutional U-Net, before spatial information is progressively recovered in the decoder.

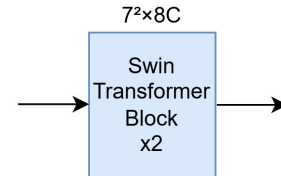


Fig. 6. The bottleneck. Two Swin Transformer Blocks at 7×7 resolution with 24 attention heads perform global self-attention over the entire compressed representation.

G. Swin Decoder and Skip Connections

The decoder (Fig. 8) mirrors the encoder with three upsampling stages. Each stage begins with a Patch Expanding layer, which doubles the spatial resolution and halves the channel dimension via a learned linear projection followed by a spatial rearrangement:

$$\text{PatchExpand} : \mathbb{R}^{B \times HW \times C} \rightarrow \mathbb{R}^{B \times 4HW \times \frac{C}{2}}. \quad (8)$$

Concretely, the linear map $\mathbb{R}^C \rightarrow \mathbb{R}^{2C}$ is applied token-wise, after which each token is reshaped into a 2×2 spatial block of $C/2$ -dimensional tokens, doubling H and W .

Skip Connection Fusion: After each Patch Expanding, the upsampled decoder features are fused with the corresponding encoder skip tensor before the Swin Transformer Blocks process the result. This ordering is critical: the transformer blocks must attend to the fused encoder-decoder representation, not to decoder features alone. Formally, let $\mathbf{D}_s$ denote the decoder tensor at stage s after upsampling, and $\mathbf{E}_s$ the matching encoder skip. The SkipConnection (Fig. 7) module computes:

$$\mathbf{S}_s = \text{GELU}(\text{LN}(\mathbf{W}_s^{1 \times 1} * [\mathbf{E}_s \parallel \mathbf{D}_s])), \quad (9)$$

where $[\cdot \parallel \cdot]$ denotes channel-wise concatenation and $\mathbf{W}_s^{1 \times 1}$ is a 1×1 convolution that projects the concatenated $2C_s$ -dimensional features back to C_s .

The concatenation-projection design preserves both encoder detail and decoder context before the attention layers refine the combined representation.

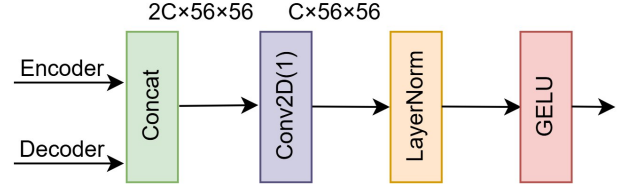


Fig. 7. The SkipConnection module. Encoder and decoder feature maps are concatenated along the channel dimension, projected to the original channel width by a 1×1 convolution, and normalised with LayerNorm followed by GELU activation.

After fusion, each decoder stage applies two Swin Transformer Blocks. The final upsampling step uses a *Final Patch Expanding* layer that performs a single $4 \times$ upsampling: a linear map $\mathbb{R}^C \rightarrow \mathbb{R}^{16C}$ followed by rearrangement into a 4×4 spatial block, recovering $56 \times 56 \rightarrow 224 \times 224$ resolution at C channels. The full decoder configuration is given in Table II.

TABLE II
DECODER STAGE CONFIGURATION ($C = 96$).

Stage	Res.	Ch.	Heads
Bottleneck input	7^2	$8C$	24
Decoder Stage 1 + skip	14^2	$4C$	12
Decoder Stage 2 + skip	28^2	$2C$	6
Decoder Stage 3 + skip	56^2	C	3
Final Patch Expanding	224^2	C	–

H. Reconstruction Block

The Swin backbone outputs a 2D feature map $\mathbf{F}_{\text{dec}} \in \mathbb{R}^{B \times C \times 224 \times 224}$. The Reconstruction Block (Fig. 9) maps this back to a full volumetric seg-

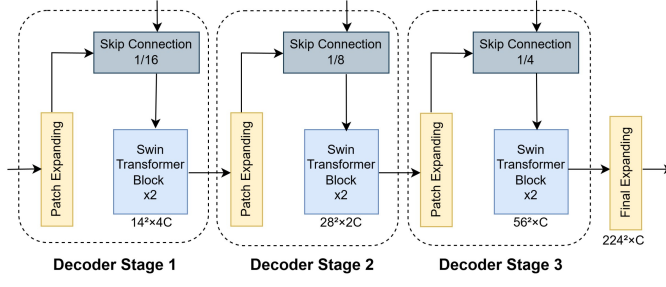


Fig. 8. The Swin decoder. At each stage, Patch Expanding first upsamples the feature map, then the SkipConnection module fuses encoder context, and finally Swin Transformer Blocks refine the combined representation.

mentation logit tensor $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times K \times D \times H \times W}$, where $K = 4$ classes, $D = 155$, $H = W = 240$.

Spatial Upsampling: Three transposed convolutions with kernel sizes 3, 5, and 11 progressively expand the spatial resolution from 224×224 back to 240×240 :

$$224 \xrightarrow{3 \times 3} 226 \xrightarrow{5 \times 5} 230 \xrightarrow{11 \times 11} 240. \quad (10)$$

Each transposed convolution is followed by Group-Norm and GELU. The channel width is reduced from C to the reconstruction hidden dimension $H_r = 64$ at the first layer and held constant thereafter.

Depth Reconstruction: A 1×1 convolution projects the H_r -channel feature map to $K \cdot D = 620$ channels:

$$\hat{\mathbf{Y}}_{\text{flat}} = \mathbf{W}_{\text{logit}} * \mathbf{x} \in \mathbb{R}^{B \times K D \times 240 \times 240}, \quad (11)$$

which is then reshaped to $\hat{\mathbf{Y}}_{3D} \in \mathbb{R}^{B \times K \times D \times 240 \times 240}$.

This design predicts each depth slice as a linear

combination of the spatial features, but provides no mechanism for inter-slice coherence.

Depth Refinement: To address this, a lightweight 3D convolution with kernel $(3, 1, 1)$ and padding $(1, 0, 0)$ is applied along the depth axis only:

$$\hat{\mathbf{Y}} = \mathbf{W}_{3d} *_{(3,1,1)} \hat{\mathbf{Y}}_{3D}. \quad (12)$$

The kernel deliberately touches no spatial neighbours (1×1 in $H \times W$) to avoid re-doing the work already performed by the Swin attention layers. Its sole purpose is to allow each depth slice to incorporate context from its immediate axial neighbours, improving coherence of the volumetric prediction for small subregions such as the enhancing tumour that span only a subset of depth slices. The spatial in-plane relationships are fully handled upstream by the Swin backbone.

I. Loss Function

We train SwinBraTS with a combined Dice–Focal loss, implemented via MONAI’s `DiceFocalLoss`. Given the model output logits $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times K \times D \times H \times W}$ and the integer ground-truth mask $\mathbf{Y}$, the loss is

$$\mathcal{L} = \lambda_{\text{Dice}} \mathcal{L}_{\text{Dice}} + \lambda_{\text{Focal}} \mathcal{L}_{\text{Focal}}, \quad \lambda_{\text{Dice}} = \lambda_{\text{Focal}} = 1. \quad (13)$$

The **Dice loss** is computed per class over the full volume after softmax normalisation, with the

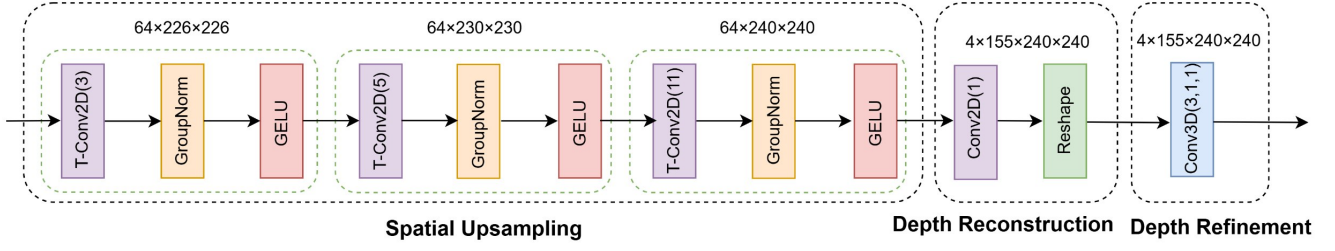


Fig. 9. The Reconstruction Block. Transposed convolutions restore spatial resolution from 224×224 to 240×240 . A 1×1 convolution expands channels to $K \cdot D = 620$, after which a reshape recovers the volumetric structure. A final Conv3d(3, 1, 1) layer introduces inter-slice context along the depth axis.

background class included:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{K} \sum_{k=1}^K \frac{2 \sum_v p_{k,v} y_{k,v}}{\sum_v p_{k,v} + \sum_v y_{k,v} + \epsilon}, \quad (14)$$

where $p_{k,v} = \text{Softmax}(\hat{Y})_{k,v}$ is the predicted probability for class k at voxel v , $y_{k,v}$ is the one-hot ground-truth, and $\epsilon = 10^{-5}$ provides numerical stability.

The **Focal loss** [13] mitigates the severe class imbalance inherent in BraTS, where background voxels dominate and the enhancing tumour region is sparse:

$$\mathcal{L}_{\text{Focal}} = -\frac{1}{KV} \sum_{k,v} (1 - p_{k,v})^\gamma y_{k,v} \log p_{k,v}, \quad \gamma = 2. \quad (15)$$

The factor $(1 - p_{k,v})^\gamma$ down-weights easy, well-classified voxels and forces the model to focus gradient signal on hard examples, primarily the rare enhancing tumour class. Labels $\{0, 1, 2, 4\}$ in the ground-truth are remapped to $\{0, 1, 2, 3\}$ before loss computation to align with the four output channels, and remapped back for evaluation.

III. EXPERIMENTAL SETUP

A. Dataset

We use the RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) 2021 Task 1 dataset [1–4, 14], which is entirely distinct from the natural image datasets used in the original Swin Transformer paper [7]. The dataset comprises 1,251 patient cases, each containing four co-registered, skull-stripped multi-parametric MRI volumes: T1-weighted (T1), contrast-enhanced T1 (T1ce, also referred to as T1Gd), T2-weighted (T2), and T2-FLAIR (FLAIR). All volumes are pre-processed to 1 mm^3 isotropic resolution with spatial dimensions $240 \times 240 \times 155$ voxels. Annotations are provided as voxel-wise segmentation masks with four semantic classes: background (label 0), necrotic tumor core and non-enhancing tumor (NCR/NET, label 1), peritumoral edema (label 2), and enhancing tumor (ET, label 4). Following BraTS evaluation convention, we also report performance on three composite regions: Whole Tumor (WT = labels 1, 2, 4), Tumor

Core (TC = labels 1, 4), and Enhancing Tumor (ET = label 4).

B. Preprocessing

All preprocessing is implemented in a custom `BraTSPreprocessor` pipeline. Each MRI volume is normalized independently using percentile-based intensity normalization: non-zero voxels (excluding skull-stripped background) are clipped to the 0.5th and 99.5th percentile range, then linearly rescaled to $[0, 1]$. This approach handles MRI intensity outliers without distorting the tissue contrast critical for tumor delineation. Each modality is saved as a separate `float32` NumPy array (`.npy`) per patient, and the segmentation mask is saved as a `uint8` array.

During data loading, each 3D volume of shape $(H, W, D) = (240, 240, 155)$ is permuted to $(D, H, W) = (155, 240, 240)$, treating the 155 axial depth slices as the channel dimension for the 2D convolutional layers in the projection and reconstruction blocks. This allows the model to process the full 3D volume in a single forward pass without slicing, while retaining the 2D nature of the Swin attention computation.

GPU-based data augmentation is applied on-the-fly during training to reduce overfitting. All augmentations are applied jointly across all four modalities and the segmentation mask to ensure spatial consistency. The augmentation pipeline consists of: ran-

dom height-axis and depth-axis flips (probability 0.4 per axis), random in-plane affine rotation up to $\pm 15^\circ$ (probability 0.2), and random multiplicative MRI bias field simulation via a low-order random polynomial field (probability 0.2, polynomial order 3). Augmentation is implemented entirely in PyTorch using `torch.nn.functional.grid_sample` on GPU, avoiding the CPU bottleneck of library-based transforms.

C. Data Splits

The dataset is split at the patient level to prevent data leakage between splits, using random splitting with a fixed seed of 42. The 1,251 patients are divided into training (75%), validation (10%), and test (15%) sets, yielding approximately 938, 125, and 188 patients respectively. All performance metrics reported in Section IV are computed on the held-out test set.

D. Evaluation Metrics

We report the Dice Similarity Coefficient (DSC) and the 95th-percentile Hausdorff Distance (HD95) for each of the three BraTS composite regions (WT, TC, ET), following the official BraTS evaluation protocol. The DSC for region r is defined as:

$$\text{DSC}_r = \frac{2|P_r \cap G_r|}{|P_r| + |G_r|}, \quad (16)$$

where P_r and G_r denote the predicted and ground truth binary masks for region r . HD95 measures the

95th percentile of the symmetric surface distance between prediction and ground truth, providing a measure of boundary accuracy that is robust to small outlier regions. Mean DSC and mean HD95 are reported as the unweighted average across the three regions.

E. Training Details

All models are trained for 105 epochs with a linear learning rate warmup over the first 5 epochs, starting from zero and increasing linearly to the peak learning rate of 2×10^{-4} . Following warmup, the learning rate is decayed using a cosine annealing schedule [15] over the remaining 100 epochs, with a minimum learning rate floor of 1×10^{-6} to prevent the loss instability that arises when the learning rate decays completely to zero.

Optimization: We use AdamW [16] as the optimizer, which decouples weight decay regularization from the gradient update step. AdamW is particularly well-suited to Transformer architectures, where the non-uniform gradient magnitudes across attention and MLP layers make hand-tuning SGD momentum impractical. A default weight decay of 1×10^{-2} is applied.

The learning rate schedule combines two phases. During the first 5 warmup epochs, the learning rate increases linearly from 0 to the peak value of 2×10^{-4} . This prevents the large gradient updates that occur at the start of training when the model

weights are randomly initialized, which can destabilize the attention layers before they have learned meaningful representations. Following warmup, a cosine annealing schedule decays the learning rate over the remaining 100 epochs according to:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{t}{T_{\max}}\pi\right) \right), \quad (17)$$

where $\eta_{\max} = 2 \times 10^{-4}$, $\eta_{\min} = 1 \times 10^{-6}$, $T_{\max} = 100$, and t is the current epoch after warmup. The cosine shape ensures the learning rate decays smoothly rather than abruptly, allowing the model to settle into a sharp minimum gradually. The floor $\eta_{\min}$ prevents the rate from reaching zero, which in our preliminary runs caused a loss spike near the end of training due to numerical instability in the AMP scaler.

The combined Dice–Focal loss described in Section II-I is used throughout training, with equal weighting $\lambda_{\text{Dice}} = \lambda_{\text{Focal}} = 1.0$ and focal parameter $\gamma = 2.0$.

Training is performed with automatic mixed precision (AMP) [17] using `torch.cuda.amp`, reducing memory consumption and accelerating forward and backward passes without loss of numerical stability. The batch size is 4, constrained by the memory footprint of the 3D volumes resolution. An early stopping criterion monitors the validation mean Dice score with a patience of 15 epochs

and a minimum improvement threshold of 10^{-3} , restoring the best checkpoint upon termination. The best validation mean Dice achieved during training was 0.6665, occurring at epoch 89.

A summary of all training hyperparameters is provided in Table III.

TABLE III
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Optimizer	AdamW
Peak learning rate	2×10^{-4}
LR schedule	Cosine annealing
LR floor ($\eta_{\min}$)	1×10^{-6}
Warmup epochs	5
Cosine annealing epochs	100
Total epochs	105
Batch size	4
Loss	Dice + Focal ($\gamma = 2.0$)
Loss weights	$\lambda_{\text{Dice}} = \lambda_{\text{Focal}} = 1.0$
Precision	AMP
Early stopping patience	15 epochs
Base channel C	24
Embed dim	48
Window size M	7
Patch size	4
Projection hidden dim	16
Reconstruction hidden dim	64

F. Hardware and Software

All experiments were conducted on a single NVIDIA H100 GPU provided by the Digital Research Alliance of Canada (DRAC). The model is implemented in PyTorch, using the MONAI library for the loss function and evaluation metrics, and TorchIO for dataset management. The full training

run of 105 epochs completed in approximately 10–15 hours. Preprocessing was performed on CPU and outputs were cached as .npy files prior to training; all data augmentation was performed on-device in GPU memory to avoid CPU bottlenecks.

IV. RESULTS AND ANALYSIS

A. Quantitative Results

Table IV reports the test set performance of SwinBraTS with the best performing configuration ($C = 24$) on the three BraTS composite regions. The model achieves a mean DSC of 0.6645 and a mean HD95 of 11.3794 mm across all sub-regions.

TABLE IV
TEST SET PERFORMANCE OF SWINBRATS ($C = 24$)

Region	DSC	HD95 (mm)
Whole Tumor (WT)	0.7761	11.2998
Tumor Core (TC)	0.6788	11.7750
Enhancing Tumor (ET)	0.5384	11.0633
Mean	0.6645	11.3794

Performance varies substantially across the three sub-regions. Whole Tumor achieves the highest DSC of 0.776, reflecting the relative ease of segmenting the combined tumor mass where all four modalities provide consistent contrast. Tumor Core drops to 0.679, as the core comprises only the necrotic and enhancing sub-regions, which are smaller and more variable in shape. Enhancing Tumor scores the lowest at 0.538, consistent with its

small and spatially irregular nature; the enhancing region is known to be the most challenging sub-region in BraTS benchmarks. HD95 scores are broadly consistent across regions, ranging from 11.0633 mm to 11.7750 mm, indicating that segmentation boundaries are reasonably well-localised but imprecise for small sub-region edges.

B. Ablation Study

To assess the sensitivity of SwinBraTS to model capacity, we trained five variants with base channel widths $C \in \{24, 48, 72, 96, 120\}$, holding all other hyperparameters constant. Table V reports test set performance for each variant.

TABLE V
ABLATION STUDY: EFFECT OF BASE CHANNEL WIDTH C

C	DSC WT	DSC TC	DSC ET	Mean DSC	#Params (M)
24	0.7761	0.6788	0.5384	0.6645	3.62
48	0.7570	0.6849	0.5427	0.6615	8.73
72	0.7513	0.6798	0.5373	0.6562	17.22
96	0.7536	0.6794	0.5293	0.6541	29.10
120	0.7619	0.6758	0.5402	0.6593	44.36

The results reveal a striking finding: the smallest model ($C = 24$) achieves the highest mean DSC of 0.6645, marginally outperforming $C = 48$ (0.6615) and $C = 120$ (0.6593). The differences across all five variants are small, within 1 percentage point, and no monotonic relationship between capacity and performance is observed. This plateau behaviour strongly suggests that the model is operating near

the capacity ceiling supported by the BraTS 2021 training set of approximately 938 patients. Larger models do not yield better generalisation because the available training data is insufficient to exploit the additional parameters. This finding is consistent with our motivation for scaling down from the original Swin-T design, and indicates that further gains from architecture scaling alone are unlikely without a corresponding increase in training data.

It is also notable that HD95 slightly favours smaller models: $C = 24$ achieves a mean HD95 of 11.38 mm compared to 12.23 mm for $C = 96$, suggesting that smaller models produce marginally tighter boundary predictions on this dataset.

To further assess the impact of data augmentation, we trained an additional variant at $C = 72$ without any augmentation, keeping all other hyperparameters identical. Table VI reports the results. Augmentation improves mean Dice from 0.6415 to 0.6562, with the largest gains on Tumor Core (+2.4pp) and Enhancing Tumor (+2.0pp) - precisely the smaller, rarer subregions where the model benefits most from seeing additional variety during training. HD95 scores are largely unchanged between the two settings, suggesting augmentation improves segmentation overlap without substantially sharpening boundary precision. Combined with the C ablation finding, this points to data augmentation as a more effective lever than model scaling for

improving performance under limited training data.

C. Qualitative Results

Fig. 10 shows the best and worst predictions on the test set, inferred by the C=24 model and ranked by mean Dice score. Each row displays the FLAIR modality, the T1ce modality, the ground truth segmentation, the model prediction, and an overlay of the prediction on the scan. The three segmentation classes are necrotic tumor core (blue), edema (green), and enhancing tumor (red).

The top row shows the best case (mean Dice 0.8551). The model accurately delineates all three subregions, with the predicted mask closely matching the ground truth in both shape and extent. The enhancing tumor boundary in particular is well-captured, which is notable given that this subregion is consistently the most challenging across the dataset.

The bottom row shows the worst case (mean Dice 0.1480). The tumor presents as a diffuse, irregular lesion, and the model produces a fragmented prediction that captures only a small portion of the edema region while failing almost entirely on the necrotic core. This failure case highlights the model’s sensitivity to atypical tumor morphologies, likely compounded by the limited diversity of training examples at the extreme end of the difficulty distribution.

D. Discussion

SwinBraTS achieves a mean DSC of 0.6645, demonstrating that the hierarchical shifted-window attention mechanism of the Swin Transformer transfers meaningfully to multi-modal MRI segmentation despite the substantial domain gap from the model’s natural image origins. The Whole Tumor region, which encompasses the full extent of the lesion across all sub-regions, is segmented most reliably, while the Enhancing Tumor, the smallest and most clinically significant sub-region, remains the most challenging. This pattern mirrors findings in the broader BraTS literature, where enhancing tumour performance consistently lags behind whole tumour due to class imbalance and the small spatial extent of the region.

The ablation study reveals that model capacity has minimal impact on performance within the range tested. This is an informative negative result: it implies that the current bottleneck is not architectural expressiveness but data quantity. With only 938 training patients, the model saturates quickly regardless of channel width. This contrasts with the original Swin-T setting, where increasing C from 96 to larger values yields consistent gains on ADE20K with 25,000 training images [7]. The domain shift thus introduces not only a distributional change but also a data scarcity constraint that fundamentally alters how the model should be scaled.

TABLE VI
ABLATION STUDY COMPARING MODELS TRAINED WITH AND WITHOUT DATA AUGMENTATION. DICE SCORES AND HD95 ARE REPORTED FOR WHOLE TUMOR (WT), TUMOR CORE (TC), AND ENHANCING TUMOR (ET).

Setting	Dice WT	Dice TC	Dice ET	Mean Dice	HD95 WT	HD95 TC	HD95 ET	Mean HD95
No Aug	0.7514	0.6554	0.5177	0.6415	12.67	12.56	11.89	12.37
Aug	0.7513	0.6798	0.5373	0.6562	13.42	11.96	11.68	12.35

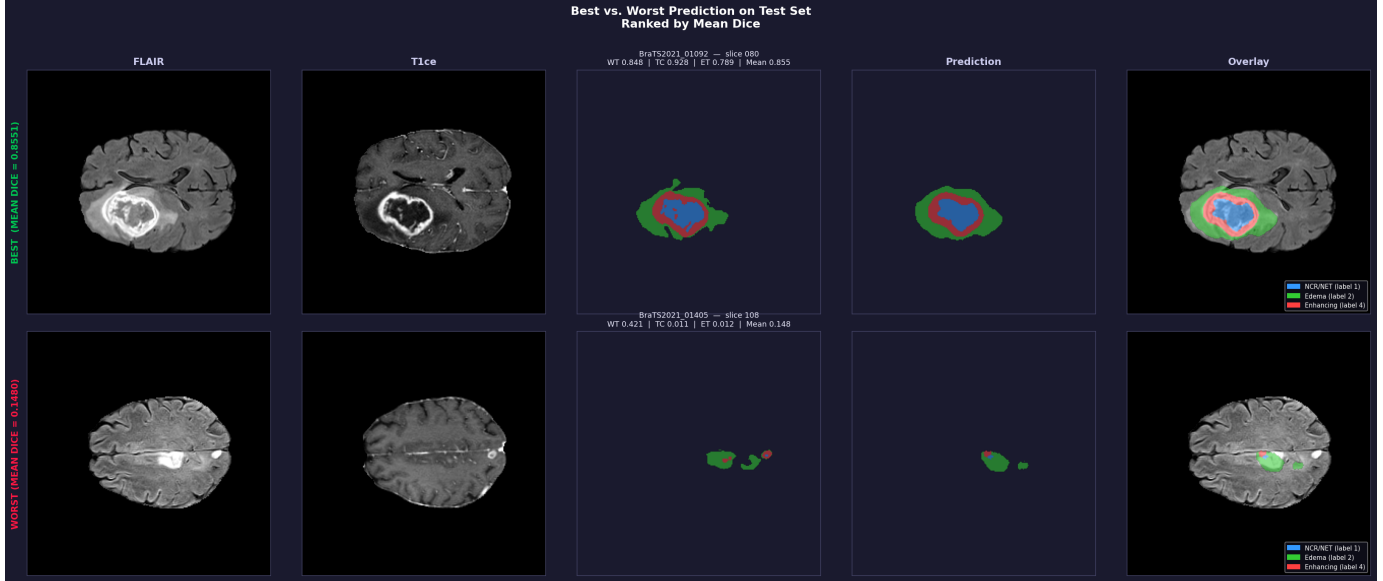


Fig. 10. Best and worst predictions ($C=24$) on the test set ranked by mean Dice score. Each row shows (left to right): FLAIR scan, T1ce scan, ground truth segmentation, model prediction, and overlay. Top row: best case (mean Dice 0.855). Bottom row: worst case (mean Dice 0.148). Colours denote necrotic core (blue), edema (green), and enhancing tumor (red).

One limitation of the current approach is the depth-as-channel treatment of the MRI volume. By permuting the 155 axial slices into the channel dimension, the model processes the entire volume in a single 2D forward pass, which is computationally efficient but discards the volumetric continuity of the data during the core attention computation. The lightweight Conv3d depth refinement layer in the reconstruction block partially compensates for this, but cannot recover inter-slice context that was never encoded by the encoder. This is reflected in

the HD95 scores: while segmentation overlap is reasonable, boundary precision along the depth axis is limited.

V. EQUITY, DIVERSITY, AND INCLUSION

The long-term survival of glioma patients is highly dependent on the composition of the tumor. Prognosis can improve through early and accurate analysis, but this is further dependent on the quality of the MRI scans. High-income countries (HIC) have access to advanced MRI technology, and have seen decreasing mortality rates each year. Whereas,

low- and middle-income countries (LMIC) such as those in Sub-Saharan Africa have disproportionately lower survival rates [18]. This is due to many factors, but at least in part to lower-resolution images and a lack of experts to diagnose them. While the BraTS 2021 dataset is comprehensive for its kind, almost all data is collected from institutions of HIC in the Global North [1]. Though we have incorporated data augmentations to increase the diversity of the data, we recognize that our model may not generalize well for underserved populations that lack access to high-resolution data.

The tools and infrastructure required to run SwinBraTS, including DRAC HPC access and substantial GPU memory, can also present barriers to adoption in under-resourced clinical setting, where advanced imaging technology may be unavailable. During training, we used optimizations such as AMP [17] to improve our model’s training speed, as well as custom data augmentations over pre-built library implementations that showed significantly slower loading times.

Future work could further address these barriers through model compression, efficient inference pipelines, and partnerships with institutions that serve diverse patient populations. Most importantly any clinical deployment of automated segmentation tools should be preceded by prospective evaluation across diverse and representative patient cohorts,

with performance reported in a disaggregated format by relevant demographic subgroups.

VI. CONCLUSION AND FUTURE WORK

This work presented SwinBraTS, an adaptation of the Swin Transformer to multi-modal brain tumor segmentation on the BraTS 2021 dataset. The domain shift from the model’s original natural image setting to four-channel multi-parametric MRI required four targeted architectural modifications: a modality projection block to bridge the input dimensionality gap, a U-Net-style decoder with skip connections to recover spatial context lost during encoding, a reconstruction block to restore full volumetric resolution, and a scaled-down model capacity suited to the limited training set. Trained from scratch on 1,251 patient cases without pre-training, SwinBraTS achieved a mean DSC of 0.6645 across the three BraTS composite regions, demonstrating that the hierarchical shifted-window attention mechanism transfers meaningfully to the medical imaging domain under these constraints.

While a mean DSC of 0.6645 demonstrates the viability of the approach, it falls short of state-of-the-art methods that benefit from large-scale pre-training and purpose-built medical architectures. The most significant limiting factor is dataset size. BraTS 2021 contains 1,251 cases, which is modest for a model of this complexity, and the additional training split constrains the depth of representation

the model can learn. This is reflected in the ablation study, where smaller values of the base channel width C performed comparably better to larger ones, suggesting the model is operating near the capacity ceiling supported by the available data.

The most direct improvement would be to increase training data volume by incorporating additional public brain MRI collections beyond BraTS 2021. In particular, datasets explicitly sourced from underserved populations such as the BraTS Africa dataset [18], further exposes the model to demographic and visual diversity. Pre-training on large-scale medical imaging datasets, or leveraging foundation models, represents a more targeted path to improving performance. Extending the architecture to fully 3D volumetric processing and reducing the computational footprint through model compression techniques are also potential next steps toward a more capable and accessible system.

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